

Aldrick Victor Tadeo Arellano

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Education

Bachelors of Science, Robotics and Digital Systems Engineering <i>Monterrey Institute of Technology and Higher Education (ITESM)</i> • Academic Excellence Scholarship recipient.	2022 — 2026 GPA: 4.0/4.0
Technical Degree, Mechatronics <i>CBTis #118 Technical High School</i>	2019 — 2022 GPA: 4.0/4.0

Skills

Programming: • Python • C/C++ • C# • JavaScript • MATLAB • VHDL	Technologies: • ROS2 • Linux • Git/GitHub • Docker • Machine Learning • Computer Vision	CAD/CAE: • SolidWorks • Autodesk Fusion • KiCAD • Siemens NX • AutoCAD	Languages: • English (Native) • Spanish (Native) • Japanese (Basic)	Soft Skills: • Critical Thinking • Adaptability • Leadership • Time Management Methodologies: • Agile/Scrum
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Work Experience

Software Engineer Intern, GE Aerospace • Designed a usage-telemetry framework for multiple internal CAD/CAE (Siemens NX) and PLM/data tools; defined event model + PostgreSQL schema for hundreds of daily users. • Integrated telemetry into engineering toolchains via a C#/.NET service wrapper, sending events through REST + internal service calls. • Developed NX Open plugins to employ NX workflows with internal tools and capture usage/error data with minimal impact to engineers. • Built Grafana dashboards to track adoption, workflow bottlenecks, and reliability trends across CAD/PLM tooling.	Nov 2025 — Present
Telemetry Development Division Lead & Founder, Electrum Performance Racing Team • Founded and led the Telemetry Division, building a robust real-time data acquisition system designed for high-stress environments. • Developed failsafe telemetry solutions using serial communication protocols, reducing transmission failures by 15%. • Improved cross-team collaboration by implementing Fusion 360's cloud-based version control, cutting design iteration times by 10%. • Directed a cost optimization initiative, reducing manufacturing and component expenses by 35% while maintaining system reliability. • Engineered firmware components to streamline workflows and enhance data accuracy across multiple vehicle subsystems.	Jun 2024 — Oct 2025

Projects

Robotic Design & Software Lead, IEEE Robosoft Competition 2026 • Designed a soft robotic gripper for strawberry harvesting, including the tool, control software, and circuitry; optimized through iterative design to reduce material costs and enhance flexibility. • Developed IoT-based crop monitoring hardware and software for greenhouse environments, improving real-time tracking of crop health. • Built an AI vision system for in-field strawberry classification, optimized for deployment on both remote servers and edge devices. • Created an object localization system using an in-hand RGB-D camera to guide robotic tool positioning with precision	Aug 2025 — Present
AI Navigation Systems Developer, Manchester Robotics & Tec de Monterrey • Developed autonomous navigation and control systems on the MCR2 Puzzlebot platform, enabling robust mobile robotics operation. • Implemented traffic sign and traffic light recognition systems, improving reliability, accuracy, and environmental adaptability by 33%. • Enhanced navigation and positioning precision by fusing visual and internal sensors for improved real-world performance. • Built vision redundancy features, reducing false positives and lowering detection/decision error rates by 15%. • Redesigned the ROS2 node-topic architecture, optimizing data throughput and minimizing system failure points.	Aug 2025 — Present

Certificates

Fundamentals of Deep Learning, NVIDIA Credential ID: 1tO0Ys3ITkGJkXM3sgBKrQ
Generative AI with Diffusion Models, NVIDIA Credential ID: TauXuWfURMOBYNutOVkopw
OpenCV Bootcamp, OpenCV University Credential ID: 65a25e083f50497dba5f5538026087
UR e-Series Tracks (Core, Pro, Application), Universal Robots